

JavaScript Loops: The Easy Guide (Numbers Only)

Think of a loop as a way to tell the computer, *"Repeat this math action until I tell you to stop."* It saves you from writing the same code over and over again.

1. The for Loop (The "Counted" Loop)

Use a for loop when you **know exactly how many times** you want to run the code before you even start (for example, exactly 10 times).

Part	Description
Initialization	Sets your starting number (creates a counter variable).
Condition	The loop keeps going as long as this math statement is true.
Increment/Decrement	Changes the counter number at the end of each turn so the loop eventually stops.

The Code Anatomy:

```
for (let i = 1; i <= 5; i++) {  
  console.log(i);  
}
```

2. The while Loop (The "Conditional" Loop)

Use a while loop when you **don't know** ahead of time how many repetitions you will need. It runs as long as a specific condition remains true.

Example: Battery Charger

```
let batteryLevel = 80;
```

```
while (batteryLevel < 100) {  
  batteryLevel = batteryLevel + 5; // Add 5% each time  
  console.log("Battery at " + batteryLevel + "%");  
}
```

3. The do...while Loop (The "Run at Least Once" Loop)

The do...while loop is almost identical to the while loop, but with one major difference: **it runs the code first, and checks the condition second**. This guarantees your code executes *at least once*.

```
let number = 10;  
  
do {  
  console.log(number); // This runs once!  
  number = number + 1;  
} while (number < 5); // The condition is immediately false
```

🌟 Star Pattern Examples (Nested Loops)

To print shapes on your screen, you put a loop *inside* another loop.

- The **outer loop** handles the rows (moving down the screen).
- The **inner loop** handles printing the items across that specific row.

Example: Right-Angled Triangle

```
let totalRows = 4;  
  
for (let row = 1; row <= totalRows; row++) {  
  let pattern = "";  
  for (let col = 1; col <= row; col++) {  
    pattern = pattern + "*";  
  }  
  console.log(pattern);  
}
```

}

Assignments

1. **The Multiples Machine (for loop):** Write a program that calculates and prints the multiplication table of the number 5, starting from 5 x 1 up to 5 x 10.
2. **Evens Only (while loop):** Start a counter at 2. Use a while loop to print all even numbers up to 20.
3. **The Divider (do...while loop):** Start with a number variable set to 100. Create a do...while loop that continuously divides that number by 2 and prints the result. Stop the loop when the number drops below 10.
4. **The Square Grid (Nested Loop Challenge):** Use nested for loops to print a perfectly square grid of stars that is 4 rows deep and 4 columns wide.